Exercise 4

SQL JOIN exercises

Join students & grades to display only students who have grades

Students

Students_id	Student_name
1	Alice
2	Bob
3	Charlie

Grade

Student_id	Grade
2	B
3	A
4	C

Output: Student_id
Student_name
Grade

```
SELECT Student_id,
       Student_name,
       Grade
FROM Students
INNER JOIN Grades on student_id = g.student_id;
```

Student_id	Student name	Grade
2	Bob	B
3	Charlie	A

Question 2 – LEFT JOIN

Display all employees & departments they belong to. Include employees with no department

Employees

emp_id	emp_name
1	John
2	Lisa
3	Mike

Departments

emp_id	dept name
2	HR
4	IT

```
SELECT emp_id
       emp_name
       dept_name
FROM employees AS e
LEFT JOIN departments d
ON e.emp_id = d.emp_id;
```

emp_id	emp_name	dept_name
1	John	NULL
2	Lisa	HR
3	Mike	NULL

Question 3 FULL OUTER JOIN

Display a complete list of products & their quantities sold
Include products with no sales & sales for unknown products

Producs

Product id	Product name
1	Laptop
2	Mouse
3	Kyboad

Sales

Product_id	Quantity
2	50
4	30

EO: Product id
Product_name

```
SELECT COALESCE(p.product_id, s.product_id) AS product_id,
       product_name
       Quantity
FROM   Products as P
FULL OUTER JOIN Sales as S
ON  p.product_id = s.product_id;
```

Product_id	Product name	Quantity
1	Laptop	NULL
2	Mouse	50
3	Keyboard	NULL
4	NULL	30

Question 4 – Left JOIN & CASE
(Logic + Filtering + Aggregates)

Objective: Display all orders & indicate whether the customer is "New or Returning"

Orders

Order_id	Customer_id	amount
1	101	500
2	102	300
3	105	0

Customers

Customer_id	Customer_name
101	Paul
102	Saral
104	Emma

EO: Order_id, customer_id, amount, customer_name
customer_type (New Customer / Returning Customer)

```
SELECT order_id
       Custumer_id
       Amount
       Customer_name
CASE   customer id IS NULL THEN 'New customer'
       ELSE 'Returningcustomer'
  END AS customer_type
FROM orders AS O
LEFT JOIN customers AS C
ON O.customers_id = C.customers_id;
```

Order_id	Customer_id	Amount	Customer_name	Customer_type
1	101	500	Paul	Returning customer
2	102	300	Sarah	Returning customer
3	105	0	NULL	New customer

Question 5 – LEFT JOIN + GROUP BY + SUM

Show total sales per region & include regions with no sales

Sales

Sale_id	Region_id	Amount
1	1	2000
2	2	3500
3	4	1000

Regions

Region_id	Region_name
1	North
2	South
3	East

Ex O: Region_id, Region name, total sales

```
SELECT region_id,
       region_name,
    COALESCE (SUM (amount), 0) AS total sales
FROM Regions AS R
LEFT JOIN  Sales AS S
ON  r.region_id = s.region_id
GROUP BY  r.region_id, r.region_name
ORDER BY  r.region_id;
```

Region_id	Region_name	Total Sales
1	North	2000
2	South	3500
3	East	0

Question 6 LEFT JOIN + CASE

Classify students based on attendance

Students

Student_id	name
1	Alice
2	Bob
3	Charlie

Attendance

Student_id	day_present
1	18
2	5
4	20

```
SELECT Student_id,
       name,
       Day_present,
CASE
   WHEN days present >=15 THEN 'Excellent'
   WHEN days present 8 and 14 THEN 'Need Improvement'
  ELSE 'Poor Attendance'
 END AS attendance_status
FROM Students as S
LEFT JOIN Attendance AS A
ON S. Student_id = A . Studentid;
```

Student_id	name	days present	Attendance Status
1	Alice	18	Excellent
2	Bob	5	Poor attendance
3	Charlie	NULL	Poor attendance

Question 7

INNER JOIN + COUNT + GROUP BY

Show number of tasks per projects. Only include projects that have have tasks

Projects

Project_id	name
1	AI chatbot
2	Website
3	Data Report

Tasks

Task_id	Project_id	Status
10	1	Complete
11	1	Pending
12	2	Complete

E.O project_id, name, task count

```
SELECT project_id
       name,
       COUNT (task_id) AS task count
FROM project AS P
INNER JOIN task as T
 ON P.project_id = t.projects_id
GROUPBY P.project_id, p.name
ORDER BY P.project_id;
```

Project_id	name	task count
1	AI chatbot	2
2	Website	1

Question 8 FULL OUTER JOIN + CASE + WHERE

Classify customers based on where they returned anything & filter by high order total

Orders			Table : Returns		
Order_id	Cust_id	Order_total	Return_id	Cust_id	Return total
1	11	120	101	11	20
2	12	250	102	14	100
3	13	180			

EO: Cust_id, order-total, return total, Return status

Filter to include only customers with total order > 100

```
SELECT Coalesce( orders.customer_id, returns.customer_id) AS cust_id,
       order total,
       return total
CASE
   WHEN returns total is NULL THEN 'No return'
   ELSE 'Returned'
  END AS Return status
FROM orders AS O
FULL OUTER JOIN Returns AS R
ON   O.cust_id = R.cus_id
WHERE COALSCE(O.roder_total, 0) >100;
```

Cust_id	Order total	Return total	Return status
11	120	20	Returned
12	250	NULL	No return
13	180	NULL	No Return

Question 9 LEFT JOIN + COUNT + ORDER BY

Count how many time each use logged in

Users

User_id	Name
1	Nelson
2	Gloria
3	Steve

Logins

User_id	Login date
2	2024-06-01
2	2024-06-02
3	2024-06-03

```
SELECT user_id
       name
    COUNT ( Logins login_date) AS login count
 FROM users AS U
LEFT JOIN logins AS L
ON u.user id = l.user_id
GROUP BY u. user.id, u.name
ORDER BY login-count DESC;
```

Userid	Name	Login count
2	Gloria	2
3	Steve	1
1	Nelson	0

Question 10 LEFT JOIN + CASE + ORDER BY

Show all teaches and the subjects they teach If no subject

Teaches

Teacher_id	Teach_name
1	Mr Hlongwane
2	Ms Nddabo
3	Mr Dlamini

Subjects

Subject id	Teacher id	Subject name
1	1	Math
2	1	Science
3	4	History

EO teacher_id teacher name Subject name (No subject assigned)
order by teacher_name ASC

```
SELECT Teachers.teacher_id,
       teacher_name
       COALESCE( Subject name 'No subject') AS. Subject name
FROM teachers AS T
LEFT JOIN subjects AS S
ON T.teacher_id = S.teacher_id
ORDER BY t.teachername ASC;
```

Teacher_id	teacher_name	Subject name
3	Mr Dlamini	No subject assigned
1	Mr Hlongwane	Math
1	Mr Hlongwane	Science
2	Mr Nddabo	No subject assigned